Backward Reasoning in Isabelle (BW)

Lecture Notes Transcription

Course: Interactive Theorem Proving (7CCSACTP)

1. Context & Comparison with Forward Reasoning

→ **Last Video (Forward Reasoning):**

- Given a rule/fact with premises ($P \implies Q$), instantiate those premises using known facts to derive new results forward (Q).

→ **This Video (Backward Reasoning):**

- Reasoning from the target goal backwards to required premises using rules (`apply (rule ...)`).

2. Intuition & Mechanics of Backward Reasoning

Core Intuition:

- Suppose your target goal is P.
- If you have a lemma `lem1`: $Q \vdash P$ (or $Q \implies P$), backward reasoning replaces the goal P with the simpler/new subgoal Q.
- **In Isabelle:** Executed via the `rule` proof method.

Example Execution:

Rule (`lem1`):	$Q \vdash P$
Current Goal:	`goal : P`
Proof Method:	`apply (rule lem1)`
New Subgoal:	`goal : Q` ✓

3. Connectives, Advantages & Next Topics

- **Operator Introduction Rules:** If the goal contains a logical connective *"op"* (e.g. $\wedge, \vee, \rightarrow, \forall$), analyze/decompose that goal using the corresponding *op-introduction* rule.
- **Why Use Backward Reasoning?**
  - **Debugging:** Allows clear visibility and debugging of proof goals.
  - **Top-Down Development:** Enables structured, top-down proof construction.
- **Next Video Topic:** Proofs by substitution (`subst`).